A Three-Regime Framework for Monetary and Banking Stability

RALE, Islands of Monetary Stability and RARB

Reserve Bank of India Governor Sanjay Malhotra,
Lord Soumadeep Ghosh and the
True Imam of India Sahin Aftab Mondal
with ChatGPT (OpenAI)

Kolkata, India

Abstract

This treatise proposes an integrated framework for monetary and banking stability organized around three complementary regimes: RALE for normal conditions, Islands of Monetary Stability for abnormal conditions, and RARB for crisis conditions. The objective is to formalize regime identification, transitions, model selection, and empirical validation without imposing a single structural model on every economic state. The framework combines nonlinear equilibrium analysis, econometrics, financial risk management, statistical classification, computational reproducibility, and institutional safeguards. Its mathematical architecture is a research specification: transition thresholds, identification conditions, and predictive performance require empirical verification before operational use.

The treatise ends with "The End"

Contents

1 Introduction **2**

2 The Three-Regime Architecture **3**
2.1 Regime roles . . . 3
2.2 State-transition diagram . . . 3

3 Mathematical Structure of Monetary Stability **4**
3.1 The Islands of Monetary Stability function . . . 4
3.2 First-order sensitivities . . . 4
3.3 Exact differential identities . . . 4
3.4 Equilibrium versus dynamic stability . . . 5

4 RALE and Banking-System Dynamics **5**

5 Regime Identification and Transition Probabilities **5**
5.1 Observable state vector . . . 5
5.2 Transition matrix . . . 6
5.3 Transition rules and hysteresis . . . 6

6 Econometrics and Empirical Validation **6**
6.1 Research hypotheses . . . 6
6.2 Benchmark design . . . 7
6.3 Identification and causal interpretation . . . 7

7 Financial Planning and Policy Analysis **7**

8 Institutional, Legal, and Governance Safeguards **7**

9 Interdisciplinary Interpretation **8**
9.1 Physics and nonlinear systems . . . 8
9.2 Chemistry and biology . . . 8
9.3 Psychology and psychiatry . . . 8
9.4 Linguistics and epistemology . . . 8
9.5 Philosophy, religion, and ethics . . . 8
9.6 Political science, geopolitics, and diplomacy . . . 9
9.7 Welfare and warfare economics . . . 9

10 Computational Implementation **9**
10.1 Pseudocode . . . 9
10.2 Software and cybersecurity . . . 10

11 Algorithmic and Statistical Failure Modes **10**

12 Welfare, Decision Theory, and Policy Losses **10**

13 Research Roadmap **11**

14 Conclusion **11**

List of Figures

1 Proposed regime architecture. Arrows indicate candidate transitions, not empirically established probabilities. . . . 3

List of Tables

1 Conceptual roles of the three models . . . 3
2 Proposed comparative evaluation . . . 7
3 Illustrative regime-contingent planning . . . 7
4 Recommended implementation milestones . . . 11

1 Introduction

Monetary and banking systems operate under changing financial, institutional, and macroeconomic conditions. A model adequate during ordinary periods may become insufficient when liquidity constraints, balance-sheet deterioration, market dysfunction, or systemic contagion emerge. Financial crises can involve mutually reinforcing feedback between leverage, asset prices, funding conditions, and intermediary balance sheets [4, 5, 6].

The architecture studied here assigns three distinct roles:

1. **RALE**: the normal-regime model;
2. **Islands of Monetary Stability**: the abnormal-regime model;
3. **RARB**: the crisis-regime model.

These assignments define the intended research architecture. They do not, by themselves, establish the empirical boundaries between regimes or prove that switching models outperforms a single-model alternative.

The central research question is whether a disciplined, evidence-based combination of these models improves the description, prediction, and management of monetary and banking risk.

2 The Three-Regime Architecture

Let the regime variable be

$$Z_t \in \mathcal{Z} = \{N, A, C\}, \tag{1}$$

where N denotes normal conditions, A denotes abnormal conditions, and C denotes crisis conditions.

The selected model is

$$\mathcal{M}_t = \begin{cases} \text{RALE}, & Z_t = N, \\ \text{Islands of Monetary Stability}, & Z_t = A, \\ \text{RARB}, & Z_t = C. \end{cases} \tag{2}$$

The model-selection rule is conditional on a regime classifier. It must not be confused with a proof that the economy naturally occupies exactly three regimes. Three regimes are a proposed modeling choice whose usefulness must be tested.

2.1 Regime roles

Table 1: Conceptual roles of the three models

Regime	**Model**	**Research objective**
Normal	RALE	Characterize the monetary and banking system under ordinary operating conditions.
Abnormal	Islands of Monetary Stability	Identify and analyze configurations of stability during departures from ordinary conditions.
Crisis	RARB	Represent crisis conditions and support the analysis of severe financial stress.

The exact equations, state variables, assumptions, and stability criteria must be taken from the respective model specifications. In particular, the acronym RARB is retained without assigning it an unverified expansion.

2.2 State-transition diagram

The following diagram represents permissible candidate transitions. A direct transition from normal conditions to crisis is included because a crisis need not pass through a separately detected abnormal regime.

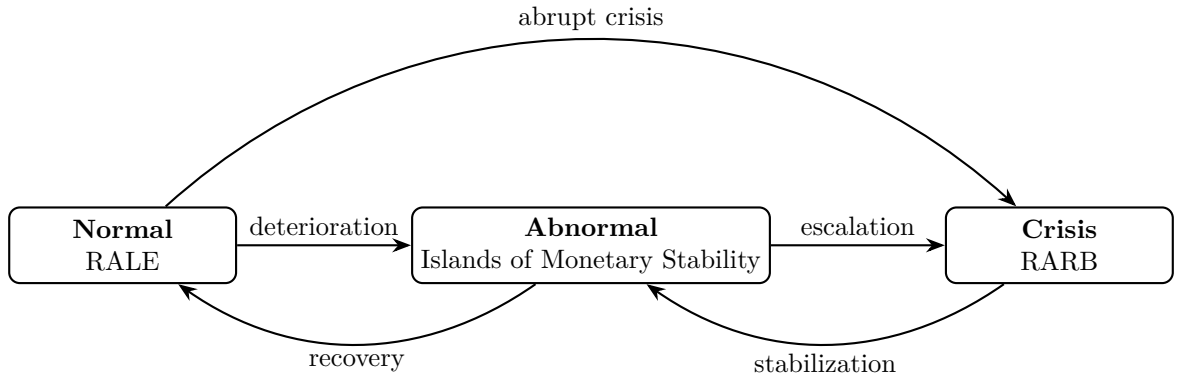


Figure 1: Proposed regime architecture. Arrows indicate candidate transitions, not empirically established probabilities.

3 Mathematical Structure of Monetary Stability

3.1 The Islands of Monetary Stability function

The function considered in the preceding analysis is

$$R(e, c, s, g) = \frac{2+e}{2-c} - \left(s + \frac{1}{g}\right)^2, \tag{3}$$

with domain

$$c \neq 2, \qquad g \neq 0. \tag{4}$$

The variables e, c, s, g retain their model-specific meanings. Their economic interpretation and admissible ranges should be defined from the underlying paper rather than inferred from notation alone.

A candidate stability boundary is the level set

$$R(e, c, s, g) = 0. \tag{5}$$

This equation defines a mathematical surface. Whether that surface corresponds to economic stability depends on the model's interpretation, equilibrium conditions, and empirical evidence.

3.2 First-order sensitivities

Differentiating with respect to each argument gives

$$\frac{\partial R}{\partial e} = \frac{1}{2-c}, \tag{6}$$

$$\frac{\partial R}{\partial c} = \frac{2+e}{(2-c)^2}, \tag{7}$$

$$\frac{\partial R}{\partial s} = -2\left(s + \frac{1}{g}\right), \tag{8}$$

$$\frac{\partial R}{\partial g} = \frac{2}{g^2}\left(s + \frac{1}{g}\right). \tag{9}$$

These derivatives describe local sensitivity. They do not independently establish dynamic stability, which generally requires analysis of the system's motion around an equilibrium.

3.3 Exact differential identities

The partial derivatives imply

$$\frac{\partial R}{\partial s} + g^2 \frac{\partial R}{\partial g} = 0, \tag{10}$$

$$\frac{\partial R}{\partial c} - (2+e)\left(\frac{\partial R}{\partial e}\right)^2 = 0. \tag{11}$$

Combining them gives

$$\frac{\partial R}{\partial c} - (2+e)\left(\frac{\partial R}{\partial e}\right)^2 + \frac{\partial R}{\partial s} + g^2 \frac{\partial R}{\partial g} = 0. \tag{12}$$

Proof. From the derivative expressions,

$$g^2 \frac{\partial R}{\partial g} = 2\left(s + \frac{1}{g}\right) = -\frac{\partial R}{\partial s}. \tag{13}$$

Also,

$$(2+e)\left(\frac{\partial R}{\partial e}\right)^2 = \frac{2+e}{(2-c)^2} = \frac{\partial R}{\partial c}. \tag{14}$$

Both identities hold throughout the stated domain. Their validity does not depend on the additional condition $R = 0$. □

These identities are algebraic consequences of the specified function. They should not be presented as independent economic laws without further interpretation.

3.4 Equilibrium versus dynamic stability

An equilibrium condition and a dynamic stability condition are different mathematical objects. For a general state vector x_t, suppose the dynamics are represented by

$$x_{t+1} = F(x_t, u_t, \varepsilon_{t+1}), \tag{15}$$

where u_t represents policy or control inputs and ε_{t+1} represents disturbances.

An equilibrium x^* in a deterministic system satisfies

$$x^* = F(x^*, u^*, 0). \tag{16}$$

Local stability depends on the behavior of perturbations around x^*. For a differentiable discrete-time system, a standard local criterion examines the eigenvalues of the Jacobian

$$J_F(x^*) = \left.\frac{\partial F}{\partial x}\right|_{x=x^*}. \tag{17}$$

If every eigenvalue has modulus strictly below one, the equilibrium is locally asymptotically stable under the usual differentiability and linearization conditions. The function R alone does not supply this Jacobian or prove that criterion.

4 RALE and Banking-System Dynamics

RALE occupies the normal-regime position in the proposed architecture. Its implementation should be documented through its actual balance-sheet identities, equilibrium equation, parameter restrictions, and numerical solution method.

A representative balance-sheet identity is

$$A_t = L_t + E_t, \tag{18}$$

where A_t denotes assets, L_t denotes liabilities, and E_t denotes equity.

If a model defines a money multiplier as assets divided by reserves, then

$$m_t = \frac{A_t}{R_t}, \qquad R_t > 0. \tag{19}$$

These identities are meaningful only under consistent definitions and units. A model implementation should test accounting identities at every simulated time step.

A documented RALE implementation has used a scalar fixed-point residual of the form

$$F(A; t) = A - R\left[1 + A - E_0(1 + A)^t\right] - X_{\text{eff}}P. \tag{20}$$

This equation is included as an implementation-specific illustration, not as a substitute for the canonical RALE specification. Its variables, units, admissible parameter ranges, and existence and uniqueness conditions must be verified against the relevant paper and code before use in a formal proof.

A bisection solver is appropriate only when a valid bracket is available and the residual is continuous on that bracket with opposite signs at its endpoints. Numerical convergence must be distinguished from economic validity.

5 Regime Identification and Transition Probabilities

5.1 Observable state vector

Let the candidate observable vector be

$$X_t = \begin{bmatrix} r_t & \pi_t & \Delta R_t & \ell_t & q_t & v_t \end{bmatrix}^{\mathsf{T}}, \tag{21}$$

where the components may represent interest rates, inflation, reserve changes, liquidity stress, credit or solvency stress, and financial-market volatility.

These are proposed candidate indicators. The final vector must match the data requirements of the three underlying models. Definitions, frequencies, units, transformations, and data vintages should be recorded in a data dictionary.

A classifier can be represented as

$$\widehat{Z}_t = \mathcal{C}(X_t; \theta), \tag{22}$$

where θ contains thresholds or estimated parameters.

The classifier should be fitted and validated using information available at the decision time. Future information must not leak into historical classifications.

5.2 Transition matrix

A first-order Markov representation uses

$$P = \begin{bmatrix} p_{NN} & p_{NA} & p_{NC} \\ p_{AN} & p_{AA} & p_{AC} \\ p_{CN} & p_{CA} & p_{CC} \end{bmatrix}, \tag{23}$$

where

$$p_{ij} = \Pr(Z_{t+1} = j \mid Z_t = i). \tag{24}$$

For a valid transition matrix,

$$p_{ij} \geq 0, \tag{25}$$

$$\sum_{j \in \{N,A,C\}} p_{ij} = 1. \tag{26}$$

These restrictions establish probabilistic coherence, not empirical correctness. The transition probabilities must be estimated or justified, with uncertainty reported.

A Markov-switching model is one possible statistical benchmark [3]. It should not be assumed to be the best classifier without comparative evaluation.

5.3 Transition rules and hysteresis

The system should define separate entry and exit conditions where appropriate. This can reduce repeated switching caused by noisy measurements.

For example, a crisis-entry indicator may be represented by

$$I_t^C = \mathbf{1}\{h_C(X_t) \geq \tau_C^{\text{in}}\}, \tag{27}$$

where h_C is a crisis-stress score and τ_C^{in} is a proposed entry threshold.

A different threshold, τ_C^{out}, may govern exit. Its numerical value must be calibrated, and the relationship between entry and exit thresholds must be specified in advance.

A robust protocol should also define the required persistence of signals, treatment of missing observations, and procedures for manual review of ambiguous cases.

6 Econometrics and Empirical Validation

6.1 Research hypotheses

The framework motivates several testable hypotheses.

H1: Predictive value. A validated three-regime system improves relevant out-of-sample forecasts relative to a single-model benchmark.

H2: Regime discrimination. The classifier distinguishes economically meaningful differences in stress and stability rather than merely reproducing arbitrary threshold labels.

H3: Early warning. The abnormal regime provides useful warning information before specified crisis outcomes, without an unacceptable false-alarm rate.

H4: Robustness. Results remain informative under reasonable alternative thresholds, data vintages, samples, and model specifications.

These are hypotheses to test, not established findings.

6.2 Benchmark design

Table 2: Proposed comparative evaluation

Design	Specification	Purpose
Benchmark A	RALE throughout the sample	Establish single-model baseline performance.
Benchmark B	RALE, Islands, and RARB selected by regime	Test the incremental value of regime-dependent modeling.
Benchmark C	Conventional statistical regime-switching model	Compare against an established alternative.

Evaluation should include forecast error, probability calibration, crisis precision and recall, false alarms, missed crises, and stability of regime assignments. Metrics should be selected according to the intended decision and outcome.

Time-series cross-validation or rolling-origin evaluation should respect chronological order. Model selection, normalization, feature selection, and threshold calibration must be performed inside the training window.

A single favorable sample is insufficient evidence of general predictive superiority.

6.3 Identification and causal interpretation

A regime classifier identifies patterns in observed data. It does not automatically identify causal effects.

Causal claims require a defensible identification strategy, such as a natural experiment, instrumental variable, structural model, or another design whose assumptions are explicitly defended. Potential confounding, reverse causality, measurement error, and policy endogeneity must be addressed [7].

7 Financial Planning and Policy Analysis

The three-regime architecture can support contingency planning by associating each regime with an analytical review and predefined decision process. It should not mechanically dictate monetary policy.

Table 3: Illustrative regime-contingent planning

Regime	Analytical focus	Planning questions
Normal	Baseline liquidity, capital, and balance-sheet dynamics	Are funding and capital buffers adequate under ordinary conditions?
Abnormal	Stability islands, sensitivity, and stress scenarios	Which configurations remain stable, and which shocks threaten those configurations?
Crisis	Solvency, contagion, funding, and systemic exposures	What are the immediate risks, legal constraints, and feasible stabilization options?

Scenario analysis should distinguish solvency from liquidity. An institution can be solvent but unable to meet immediate payments; conversely, emergency liquidity cannot by itself resolve fundamental insolvency.

Financial planning should include baseline, adverse, and severe scenarios; explicit assumptions; funding and capital constraints; sensitivity analysis; and a documented comparison of policy options. Expected losses and tail risks should be reported separately where the data permit.

A regime classification is an analytical input, not an automatic authorization to deploy public funds or impose extraordinary measures.

8 Institutional, Legal, and Governance Safeguards

The operational framework should respect applicable central-bank mandates, banking statutes, financial-stability responsibilities, due-process requirements, data-protection rules, and fiscal authorization procedures. The applicable jurisdiction must be identified before any legal conclusion is drawn.

For an Indian application, the relevant legal analysis would need to examine the Reserve Bank of India Act, 1934, the Banking Regulation Act, 1949, and other applicable legislation and regulations in their current form. Their relevance depends on the proposed intervention and institutional powers.

A responsible deployment protocol should specify:

1. the authority responsible for each decision;
2. the evidentiary basis for classifying a crisis;
3. the review and appeal mechanisms, where applicable;
4. the limits on data access and retention;
5. the documentation of model versions and decisions;
6. the procedures for correcting errors and withdrawing unsupported classifications.

Model output should inform authorized human decision-makers. Automated classification must not silently replace legal authority or accountable judgment.

9 Interdisciplinary Interpretation

The framework draws on several disciplines, but each discipline has a distinct evidentiary role.

9.1 Physics and nonlinear systems

Feedback, bifurcation, stability, and sensitivity analysis provide mathematical tools for studying transitions between dynamic regimes. Analogies with phase transitions can motivate hypotheses about abrupt changes, but economic systems are not thereby proven to obey a particular physical law.

9.2 Chemistry and biology

Concepts such as reaction networks, resilience, adaptation, and interacting subsystems can inspire useful models of financial interdependence. These are structural analogies unless a formal mapping and empirical validation are provided.

9.3 Psychology and psychiatry

Behavioral economics recognizes that expectations, bounded rationality, attention, and loss aversion can influence financial decisions [8]. Such factors may enter the model through measured behavioral variables. Psychiatric diagnoses should not be inferred from aggregate market behavior, and individual clinical categories should not be used as casual metaphors for economic regimes.

9.4 Linguistics and epistemology

A regime classifier depends on the meaning and operational definition of terms such as "stability," "abnormality," and "crisis." These terms should be defined before estimation. A change in terminology must not be mistaken for a change in the underlying economic state.

9.5 Philosophy, religion, and ethics

Epistemic humility requires distinguishing mathematical consistency from empirical truth and empirical association from causation. Ethical analysis requires consideration of distributional consequences, transparency, accountability, and the possibility that interventions benefit some groups while imposing costs on others. Religious and philosophical traditions may inform ethical reflection, but they do not substitute for empirical identification or legal authorization.

9.6 Political science, geopolitics, and diplomacy

Sovereign debt, cross-border capital flows, sanctions, conflict, and international liquidity arrangements can influence financial stability. Cross-country applications should therefore account for institutional heterogeneity, currency denomination, external financing, and geopolitical shocks.

International cooperation may improve data sharing and crisis coordination, subject to sovereignty, confidentiality, and applicable law.

9.7 Welfare and warfare economics

Policy evaluation should include output losses, unemployment, financial exclusion, fiscal costs, distributional effects, and the allocation of scarce resources. Defense expenditure and conflict-related shocks may be included as fiscal or external scenarios when supported by data. The purpose is to assess economic consequences and resilience, not to prescribe hostilities or treat war as a routine policy instrument.

10 Computational Implementation

A reproducible implementation should separate data processing, model estimation, regime selection, evaluation, and reporting. Versioned data and explicit configuration files help ensure that a result can be independently reproduced.

10.1 Pseudocode

```
INPUT:
    timestamped macro-financial data
    RALE specification
    Islands stability specification
    RARB specification
    training window and evaluation protocol

VALIDATE:
    check units, timestamps, missingness, and data provenance
    enforce chronological train/test separation
    verify model-specific parameter constraints

FIT:
    estimate permitted model parameters on training data
    calibrate regime rules using training data only
    estimate transition probabilities if justified

FOR each observation t in chronological order:
    construct X_t using information available at time t
    compute regime evidence and uncertainty

    IF crisis-entry rule is satisfied:
        Z_t = CRISIS
        evaluate RARB
    ELSE IF abnormal-regime rule is satisfied:
        Z_t = ABNORMAL
        evaluate Islands of Monetary Stability
    ELSE:
        Z_t = NORMAL
        evaluate RALE

    apply persistence and transition safeguards
    record model version, regime, inputs, and output
    do not treat model output as an automatic policy order
```

```
EVALUATE:
    calculate out-of-sample metrics
    compare against all prespecified benchmarks
    test sensitivity and calibration
    report uncertainty, failures, and limitations

OUTPUT:
    reproducible results
    transition diagnostics
    model comparison
    audit log
```

10.2 Software and cybersecurity

The implementation should validate input schemas, reject invalid numerical values, test edge cases, and log changes to model configuration. Access to sensitive banking data should follow least-privilege principles. Secrets should not be embedded in source code, and data exports should be governed by appropriate access controls.

Security testing should include dependency auditing, access control review, integrity checks, backup and recovery exercises, and incident-response procedures. Model integrity matters because tampered inputs or compromised code could alter crisis classifications.

Unit tests should verify mathematical identities, domain restrictions, accounting identities, transition-matrix normalization, and deterministic reproducibility. Integration tests should evaluate data-to-output behavior. Backtests should be separate from unit tests and should not be described as proofs of future performance.

11 Algorithmic and Statistical Failure Modes

The principal risks include:

- **Look-ahead bias:** future information contaminates historical classification.
- **Overfitting:** thresholds are optimized to one historical crisis or sample.
- **Regime ambiguity:** several regimes explain the same observations.
- **Model risk:** incorrect assumptions dominate otherwise correct computation.
- **Distribution shift:** crisis behavior differs from the training sample.
- **Feedback effects:** public knowledge of a classification changes the behavior being predicted.
- **False precision:** numerical outputs conceal weak identification or poor data quality.

An effective validation programme should report these risks explicitly rather than treating successful software execution as evidence of economic validity.

12 Welfare, Decision Theory, and Policy Losses

A classifier should be evaluated in relation to the decisions it supports. Different errors can impose different social costs. A missed crisis and a false crisis alarm need not be equally costly.

Let a denote a policy-relevant action and z the actual regime. A generic loss function is

$$L(a, z). \tag{28}$$

A decision rule may choose an action minimizing conditional expected loss:

$$a_t^* = \arg\min_{a \in \mathcal{A}} \mathbb{E}\left[L(a, Z_t) \mid \mathcal{I}_t\right], \tag{29}$$

where $\mathcal{I}_t$ denotes the information available at time t.

The action set must respect legal authority, operational constraints, and welfare considerations. The loss function should disclose how it weights financial stability, output, employment, public resources, and distributional consequences. A classifier's highest-probability label does not necessarily identify the action with the lowest social cost.

13 Research Roadmap

Table 4: Recommended implementation milestones

Stage	Deliverable	Acceptance criterion
1	Model-to-regime specification	Equations, assumptions, variables, and limitations documented.
2	Data dictionary and dataset	Provenance, units, frequency, missingness, and vintage documented.
3	Regime classifier	Rules are explicit, reproducible, and estimated without data leakage.
4	Transition model	Probabilities and persistence rules are justified and tested.
5	Benchmark evaluation	Chronological out-of-sample comparison with uncertainty reporting.
6	Governance review	Legal authority, cybersecurity, audit, and human oversight specified.
7	Stress testing	Sensitivity, distribution shift, and failure cases documented.

The first milestone should be completed before numerical transition thresholds are selected. The underlying model specifications must determine which variables and conditions are economically meaningful.

14 Conclusion

The proposed architecture assigns complementary roles to RALE, Islands of Monetary Stability, and RARB. Its contribution is not merely the use of three labels, but the attempt to connect model selection, transition rules, empirical validation, risk management, and institutional accountability within one reproducible framework.

The architecture remains a research proposition until its operational definitions, transition conditions, and predictive performance have been validated. Mathematical identities should be proved; statistical claims should be tested; legal authority should be verified; and policy recommendations should disclose uncertainty and distributional consequences.

The governing principle is that model complexity must serve explanatory and predictive adequacy. Regime transitions should be determined by documented evidence, not by intuition alone.

Three regimes, three complementary roles, one evidence-based research programme.

References

[1] Ghosh, S., and collaborators. *RALE Research and Software Repository*. Kolkata, India. Available at `https://github.com/TheRealoliGARCH/RALE`.

[2] Ghosh, S. and Mondal, S. A., *Islands of Monetary Stability*. Kolkata, India. Available at `https://github.com/TheRealoliGARCH/CollectedPapers/blob/main/Book 27.pdf`, 2–3.

[3] Hamilton, J. D. (1989). A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. *Econometrica*, 57(2), 357–384.

[4] Mishkin, F. S. (1992). Anatomy of a Financial Crisis. *Journal of Evolutionary Economics*, 2, 115–130.

[5] Brunnermeier, M. K. (2009). Deciphering the Liquidity and Credit Crunch 2007–2008. *Journal of Economic Perspectives*, 23(1), 77–100.

[6] Adrian, T., and Boyarchenko, N. (2012). Intermediary Leverage Cycles and Financial Stability. Federal Reserve Bank of New York Staff Reports.

[7] Angrist, J. D., and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

[8] Kahneman, D., and Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263–291.

[9] Engle, R. (2001). GARCH 101: The Use of ARCH/GARCH Models in Applied Econometrics. *Journal of Economic Perspectives*, 15(4), 157–168.

[10] Basel Committee on Banking Supervision (2011). *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems*. Bank for International Settlements.

[11] Government of India (1934). *Reserve Bank of India Act, 1934*. As amended. The current official text and applicable amendments should be consulted for legal analysis.

[12] Government of India (1949). *Banking Regulation Act, 1949*. As amended. The current official text and applicable amendments should be consulted for legal analysis.

Glossary

Abnormal regime
: The designated regime in which ordinary-model assumptions are considered insufficient, but the crisis-regime criteria are not met.

Calibration
: Estimation or selection of model parameters and thresholds using specified data and procedures.

Crisis regime
: The designated state in which the crisis model, RARB, is selected under the operational rules.

Equilibrium
: A state that remains unchanged under a specified dynamic system and fixed inputs.

Hysteresis
: A rule under which entry and exit thresholds differ, helping to prevent unstable switching caused by noisy observations.

Islands of Monetary Stability
: The model assigned to the abnormal regime in this framework.

Look-ahead bias
: The use of information unavailable at the historical prediction time.

RALE
: The model assigned to the normal regime. Its formal definition is determined by its canonical specification.

RARB
: The model assigned to the crisis regime. Its expansion and formal definition must be taken from the original source.

Regime classifier
: A rule or statistical model that maps observed data to a regime label or regime probability.

Transition matrix
: A matrix of conditional probabilities describing movement between regimes in a discrete-state model.

Validation
: Assessment of whether a model performs adequately for its intended use on data not used to fit it.

The End